CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 6



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic Cache Mapping

Topic Direct Mapping

Topic Tag

Topics to be Covered



- Topic: Direct Mapping
- Topic: Set Associative Mapping
- Topic: Fully Associative Mapping

[MCQ] GATE-2011

M.C.

#Q. An 8KB direct-mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following.

1 Valid bit

1 Modified bit

As many bits as the minimum needed to identify the memory block mapped in the cache.

What is the total size of memory needed at the cache controller to store meta-data (tags) for the cache?

(A) 4864 bits (B) 6144 bits

(C) 6656 bits (D) 5376 bits

[MCQ]

#Q. Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256bytes. Currently in the cache, the blocks are having tags as follows:

Block	Tag
00	10
01	01
10	11
11	01

1000 = 8
0101 = 5
1110 = 14
0111 = 7

no. of blocks = $\frac{64B}{16B} = 4$

mm add. = 8 bits

← mm block no →

Tag	CM block no.

Identify the correct statement with respect to the availability of the main memory data into cache?

a) Main memory byte number 243 present in cache
b) Main memory byte number 143 present in cache
c) Main memory byte number 43 present in cache
d) Main memory byte number 119 present in cache

$\lfloor \frac{143}{16} \rfloor = 8$

$\lfloor \frac{43}{16} \rfloor = 2$

$\lfloor \frac{119}{16} \rfloor = 7$

a) mm add. = 243

$$\text{mm block no.} = \left\lfloor \frac{243}{16} \right\rfloor = 15$$

$$\text{cm block no.} = 15 \% 4 = 3$$

$$\text{Tag} = \left\lfloor \frac{15}{4} \right\rfloor = 3$$

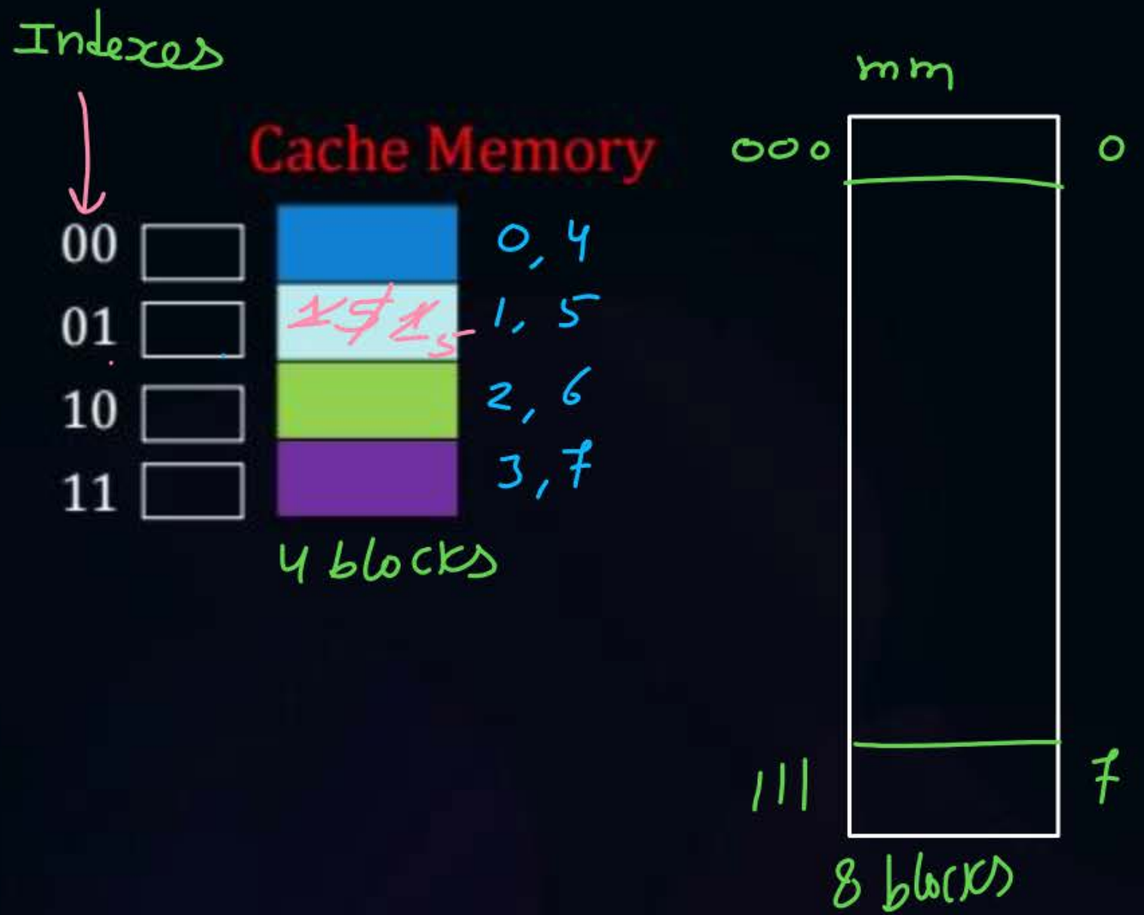
$\Downarrow$

Tag at cm block 3 is $(01)_2 = (1)_{10}$

miss



Topic : Problem With Direct Mapping



CPU Req. mm block no.

1, 5, 1, 5, 1, 5, 1, 5

miss

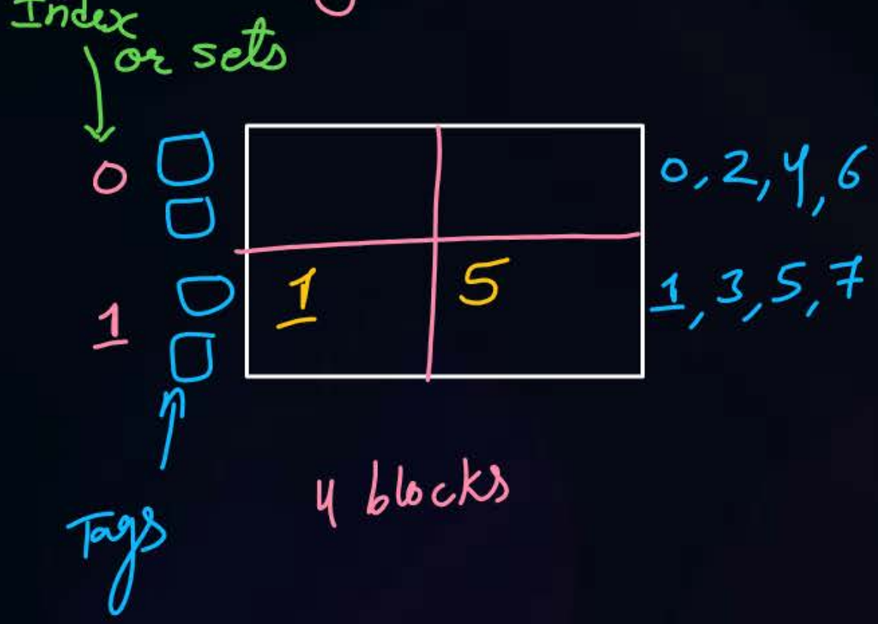
because at one index, in cache only one block among all competitors can reside.



Topic : Set Associative Mapping

organize cache in such a way that more than one block are organized under one index.

2-way set associative





CPU Req. mm blocks

1, 5, 1, 5, 1, 5, 1, 5, 1, 5

miss

miss

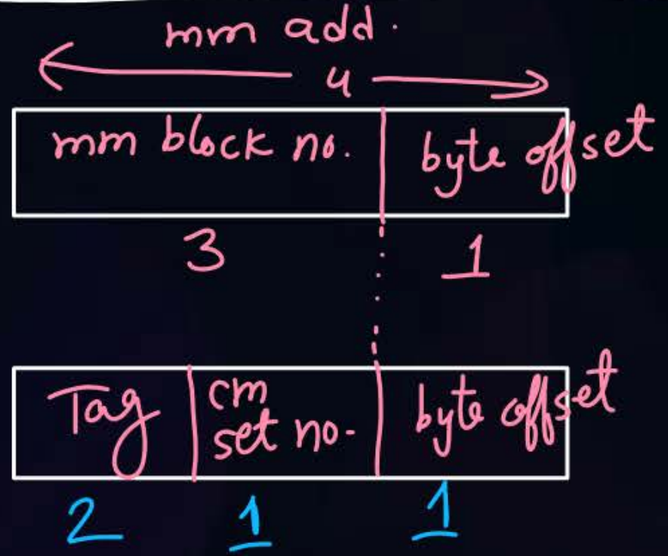
hit



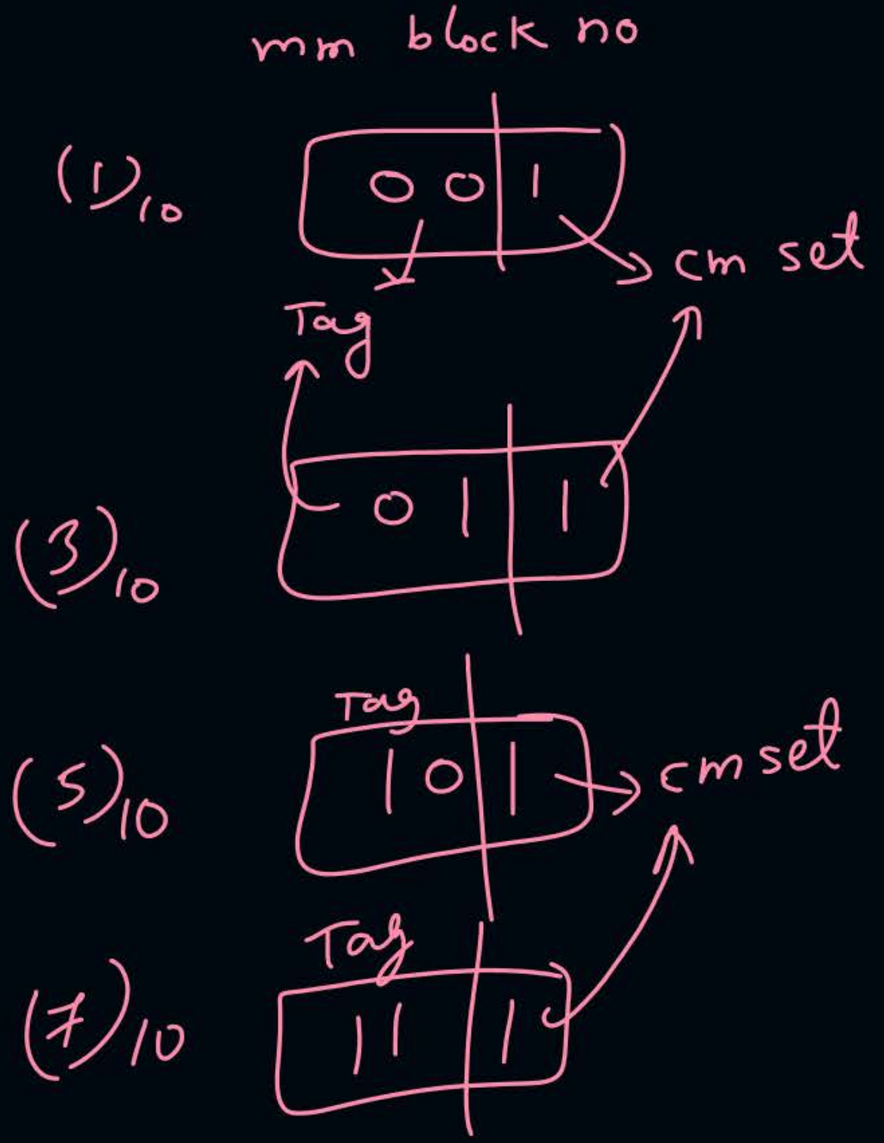
Topic : Set Associative Mapping

$$\text{cm set no.} = (\text{mm block no.}) \% \text{ no. of sets in cache}$$

$$\text{no. of sets in cache} = \frac{\text{no. of blocks in cache}}{\text{associativity}}$$



no. of bits needed for cm set no.
or
for set offset $= \log_2(\text{no. of sets in cache})$



Tag directory size

= no. of blocks * (Tag bits + extra bit)

in cache

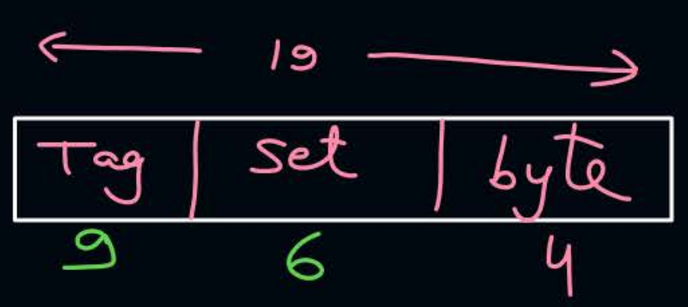
ex:-

CM size = 2KB

block size = 16B = 2^4B

2-way set ass cache

MM add. = 19 bits



Tag directory size = $2^7 * (9+0)$ bits

= $2^7 * 9$ bits

no. of blocks in cache = $\frac{2KB}{16B} = \frac{2^{11}}{2^4} = 2^7$

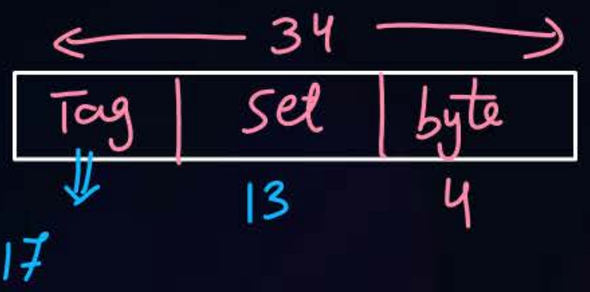
Index bits = 6 bits

no. of sets in cache = $\frac{2^7}{2} = 2^6$ ⇒ set offset = 6 bits

[NAT]

#Q. A computer has a 512Kbyte, 4-way set associative, write back data cache with block size of 16 Bytes. The processor sends 34 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit

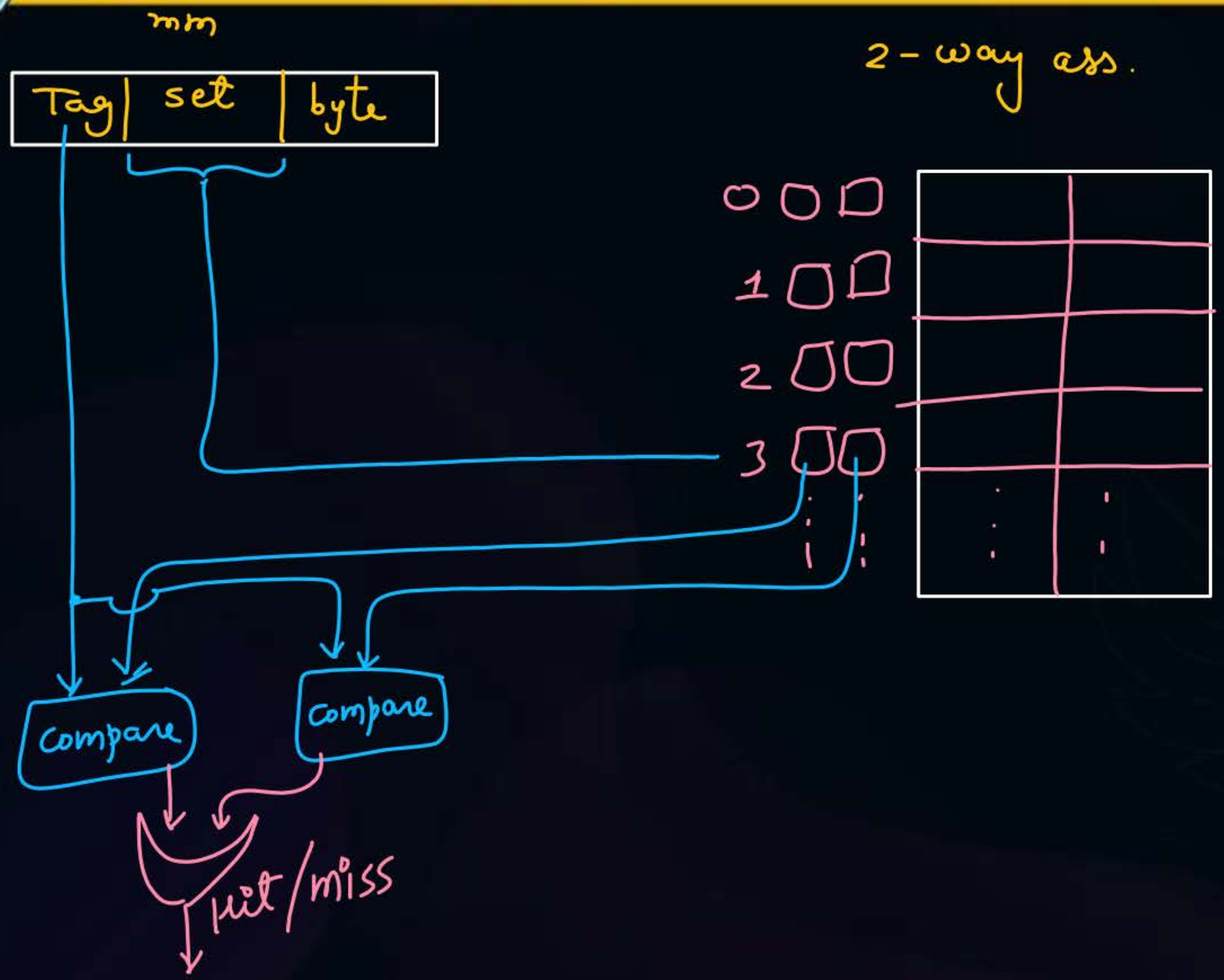
1. The number of bits in the tag field of an address is ⇒ 17 bits
2. The size of the cache tag directory is $2^{15} * (17 + 2 + 1 + 1)$ bits $= 2^{15} * 21$ bits



no. of blocks in cache $= \frac{512KB}{16B} = 2^{15}$

no. of sets in cache $= \frac{2^{15}}{4} = 2^{13}$ ⇒ set offset = 13 bits

Topic : Checking Hit/Miss in Set Associative Mapping





Topic : Tag and Index in Associative Mapping

ex :- mm add. = 16 bits

block size = 2^x B

cm size = 256 B

no. of blocks in cm = $\frac{256\,B}{2^x\,B} = 2^{8-x}$

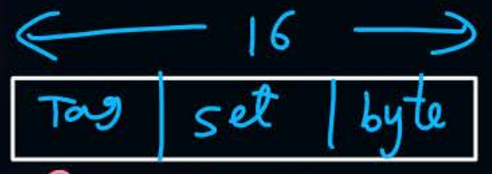
no of sets = $\frac{2^{8-x}}{2} = 2^{7-x}$

Direct mapping :-

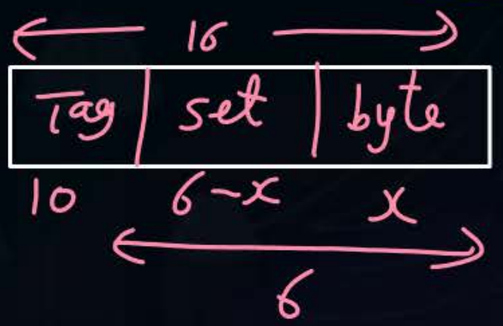


$\log_2 256 = 8$

2-way :-



4-way :-



no. of sets = $\frac{2^{8-x}}{2^2} = 2^{6-x}$

k-way set ass.

mm add.



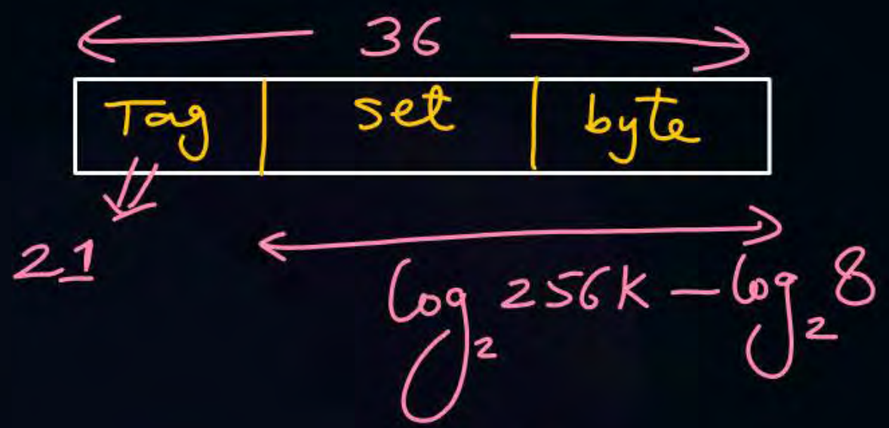
$\log(\text{cm size}) - \log_2 k$

$$\text{Tag bits} = \text{mm add} - \left[\log(\text{cm size}) - \log_2 k\right]$$

$$= \text{mm add} - \log_2(\text{cm size}) + \log_2 k$$

[NAT]

#Q. The width of the physical address on a machine is 36 bits. The width of the tag field in a 256 KB 8-way set associative cache is ___21___ bits ?



$= 18 - 3$

$= 15$



Topic : Calculating CM Set Number from MM Address

mm add.

1. binary
2. Hexadecimal
3. Decimal

mm add.

Tag	set	byte

$$\text{mm block no.} = \left\lfloor \frac{\text{mm add.}}{\text{block size}} \right\rfloor$$

$$\text{byte offset} = (\text{mm add.}) \% \text{ block size}$$

$$\text{cm set no} = (\text{mm block no.}) \% \text{ no. of sets in cache}$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of sets in cache}} \right\rfloor$$

[MCQ] GATE PYQ

↑ 2^{24} B =>

#Q. A computer system with a word length of 32 bits has a 16 MB byte- addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.

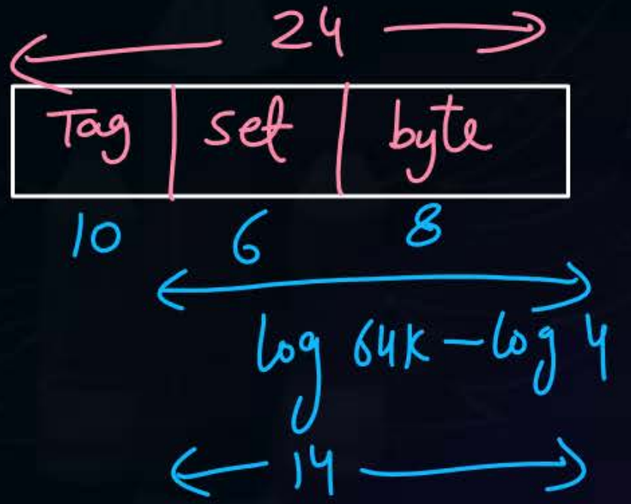
A1 = 0x42C8[illegible] 001000

A2 = 0x5468[illegible] 101000

A3 = 0x6A28[illegible] 101000

A4 = 0x5E48[illegible] 001000

Which one of the following is TRUE?



A A1 and A4 are mapped to different cache sets.

C A2 and A3 are mapped to the same cache set. ✓

A A3 and A4 are mapped to the same cache set.

C A1 and A3 are mapped to the same cache set.

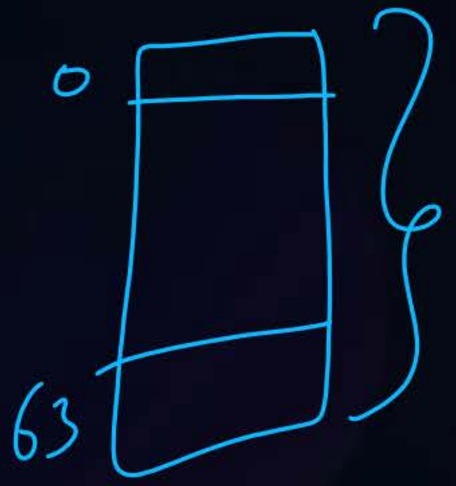


Topic : Fully Associative Mapping

ex:-
no. of blocks in cache = 64

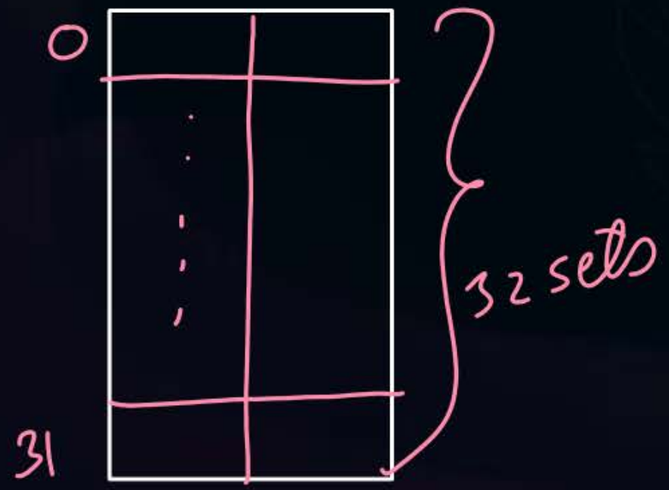
Direct :-

Tag	block	byte
y	6	x



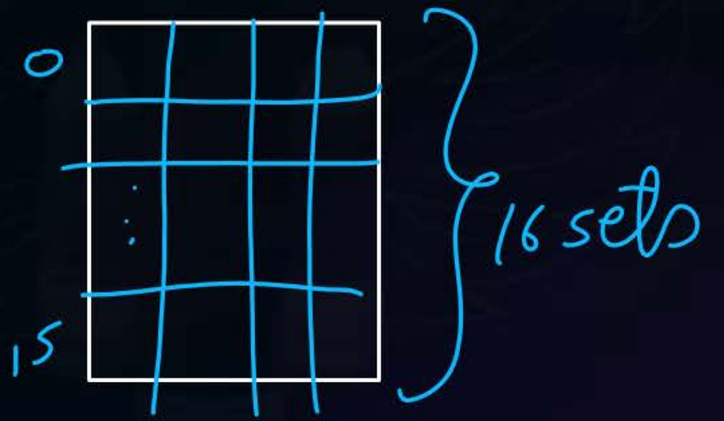
2-way :-

Tag	set	byte
y+1	5	x



4-way :-

Tag	set	byte
y+2	4	x





Topic : Fully Associative Mapping

fully ass. cache :- all blocks of cache are organized under single set.





Tag directory size = no. of blocks in cache * (Tag + extra bits)

	Index
Direct	no. of bits in cm block no.
set ass. mapping	no. of bits in cm set no.
fully ass. mapping	0 (zero)



2 mins Summary

Topic Direct Mapping

Topic Set Associative Mapping

Topic Fully Associative Mapping



Happy Learning

THANK - YOU